

NETWORK DESIGN & AUTOMATION SPECIFICATION

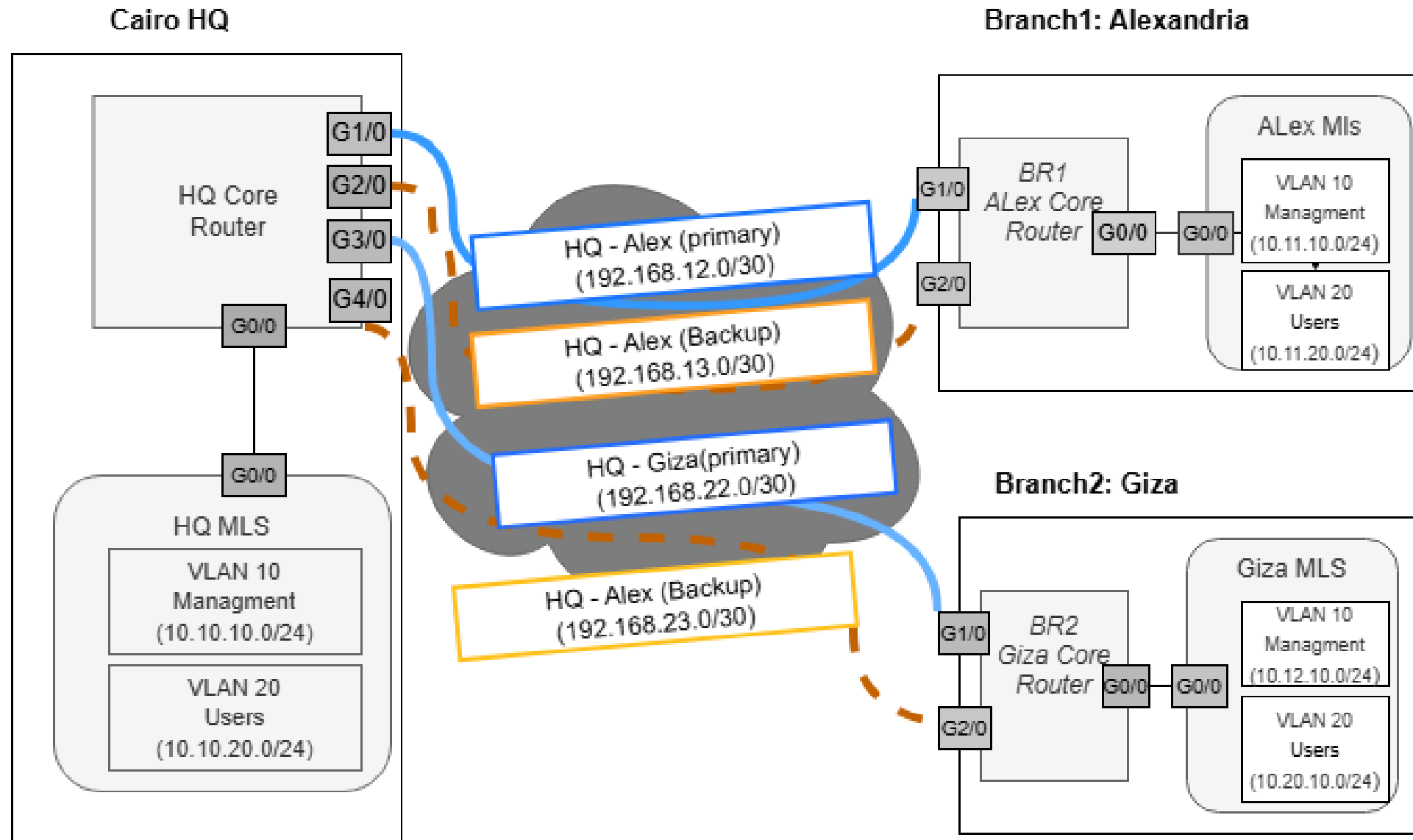
High-Availability Automated Multi-Branch WAN Architecture

Prepared For: TechLink Solutions

Target Infrastructure: Cairo HQ Core — Alexandria & Giza Regional Branches

Document Classification: Technical Design, Deployment, & Automated Telemetry Framework

❑ Network topology Diagram



❏ Master IP Addressing Table

Master IP Addressing Table					
Device name	Interface	Description	ip address	subnet mask	Default Gateway
HQ-Router		Router ID / Loopback	1.1.1.1	255.255.255.255	
	G0/0	Link to cairo ML switch	10.10.1.1	255.255.255.252	
	G1/0	Primary WAN Link to Alex	192.168.12.1	255.255.255.252	192.168.12.2
	G2/0	Backup WAN Link to Alex	192.168.13.1	255.255.255.252	192.168.13.2
	G3/0	Primary WAN Link to Giza	192.168.22.1	255.255.255.252	192.168.22.2
	G4/0	Backup WAN Link to Giza	192.168.23.1	255.255.255.252	192.168.23.2
HQ-ML-Switch	FE0/0	Routed Port to HQ Router	10.10.1.2	255.255.255.252	10.10.1.1
	FE0/1(Vlan 10)	Cairo Management SVI	10.10.10.1	255.255.255.0	
	FE1/0(Vlan 20)	Cairo Users SVI	10.10.20.1	255.255.255.0	
BR1-Alex-Router		Router ID / Loopback	2.2.2.2	255.255.255.255	
	G0/0	Link to Alex ML Switch	10.11.1.1	255.255.255.252	
	G1/0	Primary WAN Link to Cairo	192.168.12.2	255.255.255.252	192.168.12.1
	G2/0	Backup WAN Link to Cairo	192.168.13.2	255.255.255.252	192.168.13.1
BR1-ML-Switch	FE0/0	Routed Port to Alex Router	10.11.1.2	255.255.255.252	10.11.1.1
	FE0/1(Vlan 10)	Alex Management SVI	10.11.10.1	255.255.255.0	
	FE1/0(Vlan 20)	Alex Users SVI	10.11.20.1	255.255.255.0	
BR2-Giza-Router		Router ID / Loopback	3.3.3.3	255.255.255.255	
	G0/0	Link to Giza ML Switch	10.12.1.2	255.255.255.252	
	G1/0	Primary WAN Link to Cairo	192.168.22.2	255.255.255.252	192.168.22.1
	G2/0	Backup WAN Link to Cairo	192.168.23.2	255.255.255.252	192.168.23.1
BR2-ML-Switch	FE0/0	Routed Port to Alex Router	10.12.1.2	255.255.255.252	10.12.1.1
	FE0/1(Vlan 10)	Giza Management SVI	10.12.10.1	255.255.255.0	
	FE1/0(Vlan 20)	Giza Users SVI	10.12.20.1	255.255.255.0	

❑ Routing & Failover Strategy

- Protocol:** **OSPFv2** (router ospf 1, area 0) for dynamic neighbor discovery and zero manual route management.
- Traffic Flow:** Identical cost metrics enable **ECMP** (Equal-Cost Multi-Pathing). The core router splits traffic across both pipes under normal conditions to utilize all paid bandwidth.
- Failover Behavior:** If a primary link drops, OSPF drops the adjacency, flushes the bad route from the routing table, and swings 100% of traffic to the backup line.

❑ Why OSPF Over Other Protocols?

- vs. Static Routing:** Static routes require manual scripting or brittle IP SLA tracking to failover. OSPF heals itself automatically when hello packets stop, without adding complex configurations.
- vs. BGP:** BGP is built for massive internet edge or carrier networks. It introduces massive configuration overhead and slower default convergence timers that are unnecessary for a lean, three-site private WAN layout.